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Targeted mutagenesis of the P22 portal protein reveals the mechanism of signal transmission during DNA packaging

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ABSTRACT

The portal vertex in dsDNA bacteriophage serves as the site for genome encapsidation and release. In several of these viruses, efficient termination of DNA packaging has been shown to be dependent on the density of packaged DNA. The portal protein has been implicated as being part of the sensor that regulates packaging termination through DNA-dependent conformational changes during packaging. The mechanism by which DNA induces these conformational changes remains unknown. In this study, we explore how point mutants in the portal core can result in changes in genome packaging density in P22. Mutations in the portal core that subtly alter the structure or dynamics of the protein result in an increase in the amount of DNA packaged. The magnitude of the change is amino acid and location specific. Our findings suggest a mechanism wherein compression of the portal core is an essential aspect of signal transmission during packaging.

1. Introduction

For dsDNA bacteriophage and herpesviruses, virus assembly begins with the viral coat or capsid proteins polymerizing into an empty icosahedral shell, or procapsid (The Bacteriophages, 2006; Aksyuk and Rossmann, 2011; Casjens, 1985). Following procapsid assembly, the newly replicated genome is pumped into the preformed capsid by the terminase complex – a powerful molecular motor that drives the viral DNA into the capsid against concentration and pressure gradients in a highly ATP-dependent process (Aksyuk and Rossmann, 2011; Casjens, 2011). In many of these viruses, genome encapsidation is accompanied by an irreversible expansion of the capsid (Aksyuk and Rossmann, 2011; Chen et al., 2010; Guo et al., 2014; Lee et al., 2008). This expansion process is necessary to both stabilize the capsid against increasing internal pressures and to provide sufficient internal volume to accommodate the entire viral chromosome (Chen et al., 2010; Guo et al., 2014; Lee et al., 2008). Following genome packaging, the DNA is stabilized inside of the virion through the addition of the tail machinery (Aksyuk and Rossmann, 2011; Casjens, 2011; Lander et al., 2009; Olia et al., 2011). In order for the viral genome to be packaged into these pre-formed heads, the procapsids must contain an entryway. This entryway is formed through the placement of a 12-fold symmetric ring into one of the 5-fold vertices in the icosahedral lattice (Olia et al., 2011; Lebedev et al., 2007; Simpson et al., 2000; Sun et al., 2015). This insertion disrupts the strict icosahedral symmetry of the capsid, while forming a conduit through which DNA can pass (Bhardwaj et al., 2014). In all reported cases, this oligomeric ring is composed of twelve copies of a single gene product referred to as the portal or connector protein that share a common general architecture (Olia et al., 2011; Lebedev et al., 2007; Simpson et al., 2000; Sun et al., 2015; Bhardwaj et al., 2014; Agirrezabala et al., 2005; Trus et al., 2004). In addition to forming the conduit through which the DNA traverses the capsid barrier, the portal ring also serves as the binding site for the terminase complex during genome packaging and the virion-stabilizing tail machinery following capsid maturation (Casjens, 2011; Lander et al., 2009; Olia et al., 2011; Lebedev et al., 2007; Simpson et al., 2000; Hegde et al., 2012; Oliveira et al., 2010; Sun et al., 2008; Dixit et al., 2013; Viral Genome, 2005; Black, 1989).

Most of the known viruses that employ this two-step assembly mechanism encapsidate linear, dsDNA genomes (Viral Genome, 2005; Black, 1989; Casjens et al., 2011). Of these viruses, most package their genomes from concatemeric DNA substrates produced via rolling circle replication of a circularized viral chromosome (Viral Genome, 2005; Black, 1989; Casjens et al., 2011). Packaging is initiated when terminase makes a site-specific double-stranded incision in a single genome in the concatemer (Viral Genome, 2005; Black, 1989; Casjens et al., 2011). This DNA cleavage event generates the 5′ end of the chromosome that is going to be packaged. Terminase then binds to the procapsid at the portal vertex and packaging proceeds unidirectionally in a 5′ – 3′ orientation (Viral Genome, 2005; Black, 1989; Casjens

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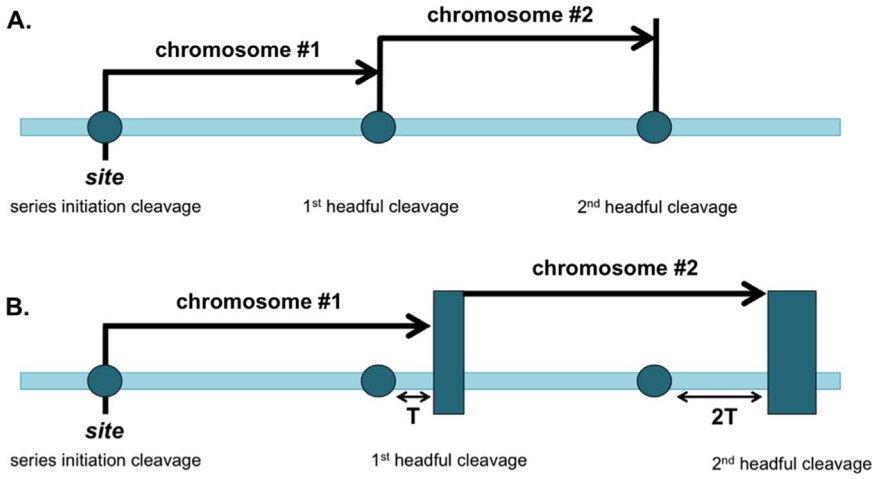

Fig. 1. Generalized packaging strategies in portal-containing viruses. (A) Schematic cartoon of the sequence-dependent packaging strategy. Packaging is initiated at a single location in the concatemer at a particular DNA sequence, denoted here as site. Packaging proceeds unidirectionally into the head until site is recognized again by the terminase complex and the DNA is cleaved. The right-hand side of the cleaved DNA becomes the 5' end of a new chromosome to be packaged into a new head. All of the packaged chromosomes in these viruses are identical in both length and nucleic acid sequence. (B) Schematic of the sequence-independent packaging strategy. Like the sequence-dependent mechanism, packaging is initiated from a particular site in the concatemer. However, packaging into the first head proceeds past the location of this specific sequence in the adjacent genome in the concatemer before packaging is terminated. The consequence of this is terminal redundancy on the two ends of the packaged chromosome, denoted in the diagram as T. Additionally, the headful mechanism has an associated uncertainty, denoted by the dark rectangles, that is absent in sequence-dependent packaging strategies. The diagrams in this figure are adapted and modified from Casjens et al. (1992).

et al., 2011).

The most apparent difference in the packaging strategies among these types of viruses is in how packaging is terminated (Casjens et al., 2011). There are two general types of termination strategies: sequence-dependent and sequence-independent (Fig. 1A and B). In sequence-dependent strategies (Fig. 1A), terminase packages DNA into a given capsid until a specific DNA sequence signals the terminase to cleave the DNA and halt packaging into that particular particle (Black, 1989; Casjens et al., 2011). In sequence-dependent strategies, all packaged chromosomes are identical in both length and gene order. In the sequence-independent packaging strategy (Fig. 1B), terminase halts packaging by cleaving the DNA when a particular amount of DNA has been packaged inside the head, regardless of the DNA sequence at that point in time (Black, 1989; Casjens et al., 2011; Casjens and Weigele). This packaging strategy results in chromosomes that are both terminally redundant and circularly permuted (Casjens et al., 2011). The terminal redundancy dictates that slightly more than one genome-length of DNA is packaged into each capsid. This ensures that there is sequence overlap on both ends of a single chromosome, though the end sequences between chromosomes differ. This facilitates circularization of the genome in the host cell cytoplasm and proper DNA replication.

In the absence of a sequence-dependent termination strategy, the only metric for packaging progress into a given head is the buildup of DNA density. Somehow, information on DNA density has to be transmitted from the interior cavity of the capsid to the terminase complex on the capsid exterior for termination to occur. The portal ring is the only component of these virus capsids that spans both the interior and exterior surfaces of the capsid. Moreover, the portal vertex serves as both the binding site for terminase and the conduit through which DNA is packaged. These facts strongly implicate the portal protein as being involved in the sensing mechanism required for packaging termination in these viruses. Indeed, point mutants in the P22 and SPP1 portal proteins have been shown to influence the extent of packaging in these systems (Isidro et al., 2004; Casjens et al., 1992). Interestingly, it has also been shown that packaging density is an important feature of termination even in phage with sequence-dependent termination mechanisms, although it does not serve as the final signal for DNA cleavage in these systems (Feiss et al., 1976; Cue and Feiss, 1997). For example, in bacteriophage λ, when less than 75% of the genome is packaged, termination does not occur even when the defined termination sequence is present (Feiss et al., 1976; Cue and Feiss, 1997).

In this study, we build on the recently published structural work showing distinct conformational states of the P22 portal protein by exploring the origin of signal transmission across the portal itself. We have used the bacteriophage P22 as the model system for our work. P22 is a tailed, dsDNA bacteriophage that packages its approximately 43 kilobase genome in a strictly headful (sequence-independent) manner (Casjens and Weigele). Because of this, changes in the DNA packaging density inside of the head can be directly related to portal protein function. This makes P22 an ideal model system for biochemically dissecting portal function during packaging.

The recently published structure of the procapsid form of the dodecameric P22 portal demonstrates that portal undergoes pronounced structural rearrangements and conformational changes during packaging (Lokareddy et al., 2016). The two principle findings of this study are that 1) the portal subunits in the procapsid form of the structure are arranged in an asymmetric manner and 2) the conformation of the individual portal protomers is distinct from that found in the mature state (Lokareddy et al., 2016). The most dramatic conformational changes were observed in the stalk domain, which comprises most of the terminase binding site, and the wing domain (Fig. 2A) (Lokareddy et al., 2016). These domains are located at opposite ends of the protein, and communication between them must be mediated by intermediate residues.

Based upon the asymmetric reconstruction of packaged P22 virions, it was suggested that pressure from DNA packaging influences the conformation of the C-terminal helical barrel (Tang et al., 2011). Because packaging density is accompanied by a proportional increase in pressure, it is likely that pressure-induced conformational changes are part of the DNA density sensing mechanism. In addition to the high resolution structural data showing distinct forms of the P22 portal protein in procapsids and virions, respectively, a recently published study on the phage φ29 identified a "continuous allosteric change" that likely occurs in the portal-terminase complex during packaging (Berndsen et al., 2015). This observation is consistent with the notion of pressure-induced changes influencing DNA packaging.

Our results demonstrate that mutations throughout the core of the protein lead to overpackaging phenotypes in P22. The extent of this phenotype varies. A series of mutations at two previously identified

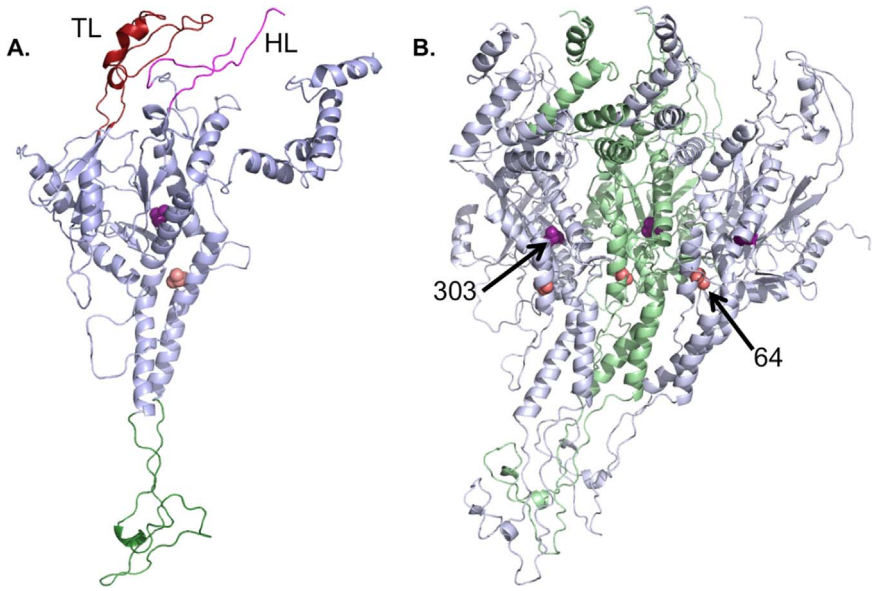

Fig. 2. Structural features of the P22 portal protein (A) Cartoon representation of a portal monomer extracted from the crystallized mature dodecamer. In this cartoon, the trigger loop (TL) is highlighted in red, the hammer loop (HL) is highlighted in purple, and the stalk is highlighted in green. These are the regions in which the most pronounced conformational changes are seen between the two portal conformations. Communication between them cannot be direct. Instead, information must be transmitted through the rest of the portal structure. (B) A trimer of portal monomers extracted from the crystallized dodecameric ring oriented with the central channel facing the viewer. The locations of residues 64 and 303 are shown in salmon and violet, respectively, and are denoted by the arrows. This cartoon shows that the positions of 64 and 303 are in the core of the protein, and that they do not appear to be involved in any known functional domains of portal or directly involved in the interaction between individual subunits. All structures were made with PDB ID: 4V4K, published in Ref. Olia et al. (2011).

sites that affect packaging in P22 result in amino acid-specific phenotypes. The chromosome sizes in these phage are unique for each mutant but lie within a continuous range of values. Mutations in the central structural motif of the portal protein, however, appear to be location-dependent, as mutations throughout this motif result in packaging phenotypes that are clustered around particular values. Biophysical characterization of a subset of these mutant proteins suggests that they result in subtle structural or dynamic changes that alter the sensitivity of the protein to packaging density. We attribute this loss of sensitivity to changes in the overall compressibility of the portal protein.

2. Results

2.1. Mutations at positions 64 and 303 affect packaging in an amino acid-specific manner

A previous study of randomly induced mutations in P22 identified two sites in the portal protein – residues 64 and 303 – at which valine to methionine substitutions result in a 4–5% overpackaging of DNA (Fig. 2B) (Casjens et al., 1992). These residues are located near the core of the protein, far removed from any important binding interface or previously identified functional domain. Given their location in the protein, it is unlikely that the observed packaging phenotypes in these mutants are the result of direct interaction with the DNA or with the terminase complex. Instead, we hypothesize that the mutations at positions 64 and 303 lead to overpackaging by affecting the transition between the procapsid and virion forms of the portal protein.

To begin our studies, we introduced a series of mutations into positions 64 and 303 with side chains both larger and smaller than the endogenous valine residues. At position 64, we introduced A, T, M, and L mutations. At position 303, we introduced A, T, M, and Y mutations. To analyze the affect of these mutations on packaging, we employed a previously described transcomplementation system to make phage particles containing the desired portal proteins (Moore and Prevelige, 2002). All of the mutants titer at levels comparable to WT phage (data not shown), indicating that all of the mutants are functional *in vivo*. In order to assess the influence of the mutations on DNA packaging, we purified the packaged chromosomes from WT and mutant phage and determined their respective sizes via analytical ultracentrifugation (Fig. 3A). All of the chromosomes sedimented as a single sharp peak with s-values between 31.5 and 32.5 S (Figs. 3A and 3B). The mass of each chromosome was calculated using a previously published scaling law (Fig. 3B).Clark et al. (1980) Shown on the plot in red, is the chromosome mass for the 64 M mutant obtained from a direct mass measurement using charge-detection mass spectrometry (Contino et al., 2013; Keifer et al., 2015; Keifer et al., 2014). The good agreement between the two techniques highlights the accuracy of the scaling law. In Fig. 3C, we plot the measured chromosome size per mutant, arranged from smaller to larger values. The chromosome size of the WT phage was found to be 43.1 kbp. This value is close to the previously reported value of 43.4 kbp determined by restriction digestion of packaged chromosomes (Casjens and Hayden, 1988). The chromosome size of the overpackaging mutant 64 M is also consistent with its previously determined value (Casjens et al., 1992). The mutant chromosomes were all found to be longer than WT, indicating that all of the mutants have an overpackaging phenotype (Fig. 3C). The degree of overpackaging, however, varies substantially. The substitution with the smallest effect on chromosome length is 64 T, with a calculated chromosome size of 43.2 kbp, just 0.1 kbp greater than the WT chromosome. Conversely, the mutation with the most pronounced packaging effect, 64 L, has a chromosome size of 46.0 kbp, or 2.9 kbp greater than WT. All of the other measured chromosome sizes fall into the 2.8 kbp range between these two end points. Interestingly, the measured chromosome sizes lie on an apparent continuum of values, with no obvious clustering at discrete packaging intervals. A plot of chromosome size vs. amino acid volume (Fig. 3D) indicates that there is no correlation between packaging phenotype and amino acid size.

2.2. The introduced mutations do not affect the fundamental packaging mechanism

Due to the termination strategy of P22, there is intrinsic variability

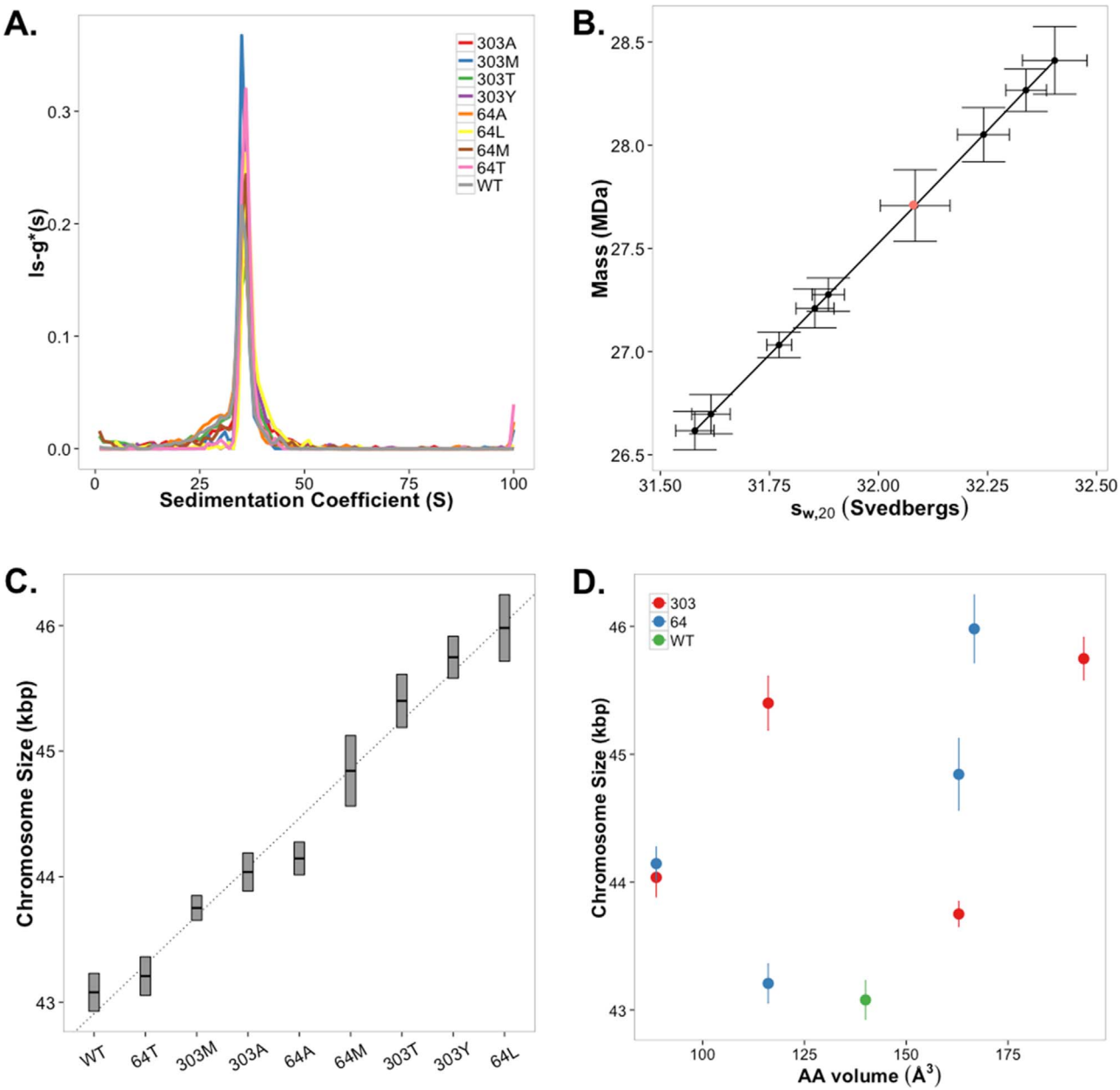

Fig. 3. Packaging phenotypes of the mutants at positions 64 and 303. (A) Example sedimentation coefficient distributions for the chromosomes purified from P22 virions containing mutant portals. All of the analyzed chromosomes show sharp, single-species distributions. (B) Plot of the calculated mass of each chromosome vs. the average s-value. From the x-axis, it can be seen that all of the mutants had chromosomes with reproducible s-values between 31.5 and 32.5 S. This corresponds to a 2 MDa range is mass values. The control point (shown in red) was calculated from direct mass measurements of the 64M phage using charge detection mass spectrometry. The fact that this point is nearly superimposable with the results from centrifugation highlight the accuracy of the scaling law. (C) Plot of average chromosome size per mutant arranged in increasing order. From this plot, it is apparent that all of the generated mutants package more DNA than the WT phage. The measured chromosome sizes for the mutants lie on an apparent continuum of values. The line in the plot is drawn to aid the eye. This feature of the data is consistent with either a continuous mechanism or a highly tuned binary switch. (D) Plot of measured chromosome size vs. amino acid volume. No correlation between chromosome size and amino acid volume was found. This lack of correlation suggests that the transition between portal conformations occurs as a continuuous mechanism.

in the chromosome lengths from individual packaging events. Previously, the range of variability in WT P22 was estimated to be ± 750 bp (Casjens and Hayden, 1988). This variability is an inherent feature of the packaging mechanism and is indicative of the manner in which the phage responds to packaging density. By calculating error surfaces at 1σ above and below the mean sedimentation coefficient for each construct, we are able to estimate the distribution of chromosome lengths in each mutant. The range of chromosome lengths for each mutant is plotted in Fig. 4A. The range of packaging lengths observed in the WT sample is 1.4 kbp, close to the previously estimated range of 1.5 kbp. The mutants had distribution widths similar to that of WT, ranging from 0.9 to 1.6 kbp. Moreover, the calculated distribution widths on either side of the mean were approximately symmetric for all constructs, indicating that the uncertainty for the WT and mutant constructs was not biased in either direction. These data suggest that the fundamental response mechanism is unaltered by the introduced mutations and that the process that we are comparing between constructs is similar.

2.3. Observed packaging phenotypes are not due to changes in particle morphology

Portal proteins have been linked to capsid morphology and maturation (Ray et al., 2009; Guo et al., 1991). To ensure that the

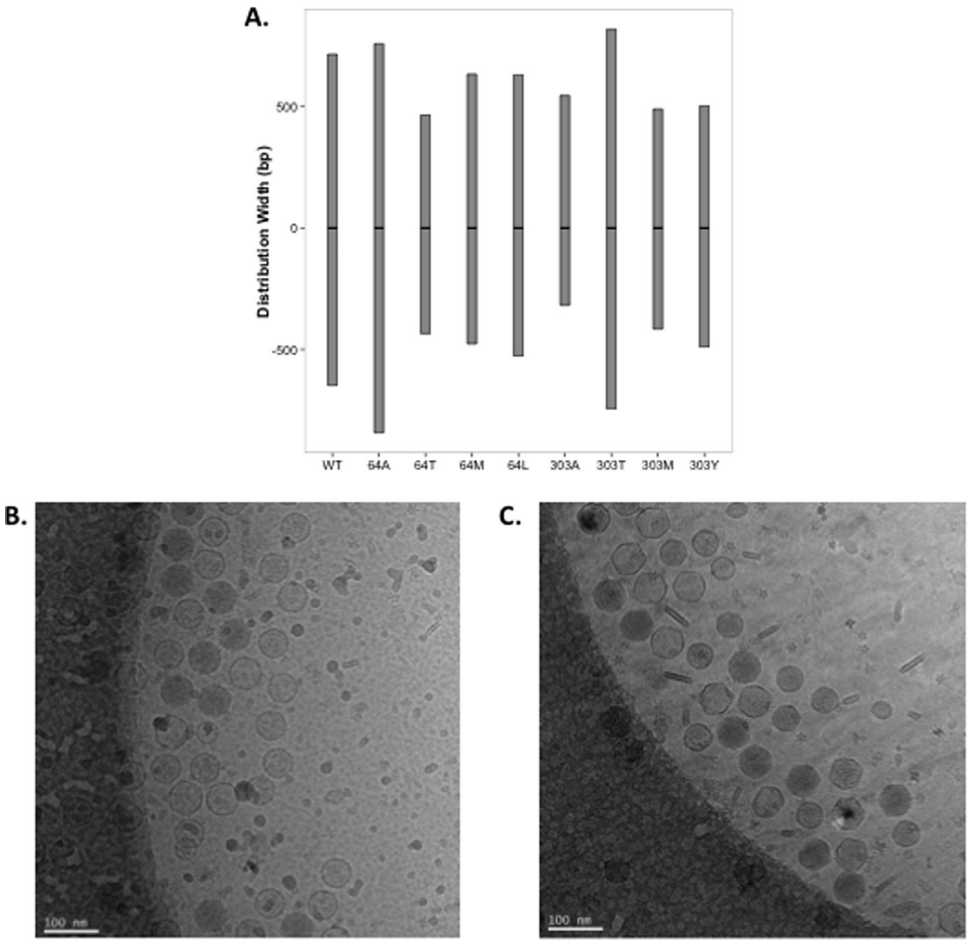

Fig. 4. Packaging error and particle morphology of overpackaging mutants. (A) Plot of the distribution width for each mutant calculated at 1-sigma. Calculation of the error surface in this way allows us to estimate the polydispersity in each sample. Previously, the WT P22 imprecision was estimated to be ± 750 bp. Our results are similar to that value. Additionally, all of the mutants have distribution widths similar to that of WT, suggesting that the fundamental packaging process is similar between the WT and mutant phage. (B) WT and (C) 64L cryo-EM micrographs of particles produced in the transcomplementation system. The particles in the 64L micrograph are indistinguishable from WT, suggesting that the portal mutants that affect packaging do so in a packaging-specific manner and that they do not alter other aspects of phage assembly or maturation.

observed packaging phenotype is not due to changes in particle morphology or does not influence other aspects of P22 maturation, we imaged crude particle preps of WT and 64 L particles produced in the transcomplementation system by cryo-EM. The 64 L mutant was chosen because it is the most extreme overpackager in our collection of mutants. This imaging strategy allows us to analyze particle morphology as well as verify that particle distribution is similar in solution. Both WT and 64 L lysates contained procapsids, empty expanded heads, and mature virions. The particle sizes of WT and 64 L phage were determined to be 58.43 (± 0.98) nm and 58.84 (± 1.09) nm, respectively (Figs. 4B and 4C). This finding is consistent with previous results that packaged chromosome length does not influence phage morphology (Casjens et al., 1992). Additionally, the overall number of procapsids, empty expanded heads, and mature virions is comparable between WT and 64 L micrographs. These data indicate that there are no mutant-specific defects in terminase or tail machinery attachment, in particle expansion, or in particle stability.

2.4. The portal core is generally involved in chromosome size determination

At the center of the portal core is a stretch of 28 amino acids (residues 105–132) that form an alpha helix connected to a beta strand by a short unstructured hinge. This group of amino acids is buttressed on either side by the structural regions containing residues 64 and 303 and lies directly on top of the back stem helix in the portal structure. Given the location of this stretch of amino acids relative to the known packaging mutants, we hypothesized that amino acid changes in this structural motif will also result in observable packaging phenotypes. To test this hypothesis, we generated a series of mutations spanning the region in question (Fig. 5A, red). Within this region, residues N112, I119, G124, W126 were mutated to alanine and residues A110 and A114 were mutated to leucine. All of the newly constructed mutants titer at levels comparable to WT (data not shown). Shown in Fig. 5B, all of these mutants resulted in an overpackaging phenotype. The magnitude of this phenotype is larger for this set of mutants than for all but two of the mutants at positions 64 and 303. The measured chromosome size of each of these mutants varied between 45.5 and 46.8 kbp (Fig. 5B). The mutants W126A and A110L both result in chromosome lengths of ~45.5 kbp. The mutants A114L, I119A, N112A, and G124A, on the other hand, all result in chromosome sizes in a very narrow range between 46.5 and 46.8 kbp. This apparent clustering at discrete packaging intervals is in stark contrast to the results obtained for the mutants at positions 64 and 303. The locations of these mutants can be seen in Fig. 5C. In this figure, each residue is colored by the extent of packaging observed. Residues 110 and 126, blue, map to adjacent locations in the protein, closer to the top of the region in question and farthest away from positions 64/303 and the stem helices. The remaining four mutants cluster together in the middle/bottom of the region and are closer in proximity to residues 64/303 in the portal structure.

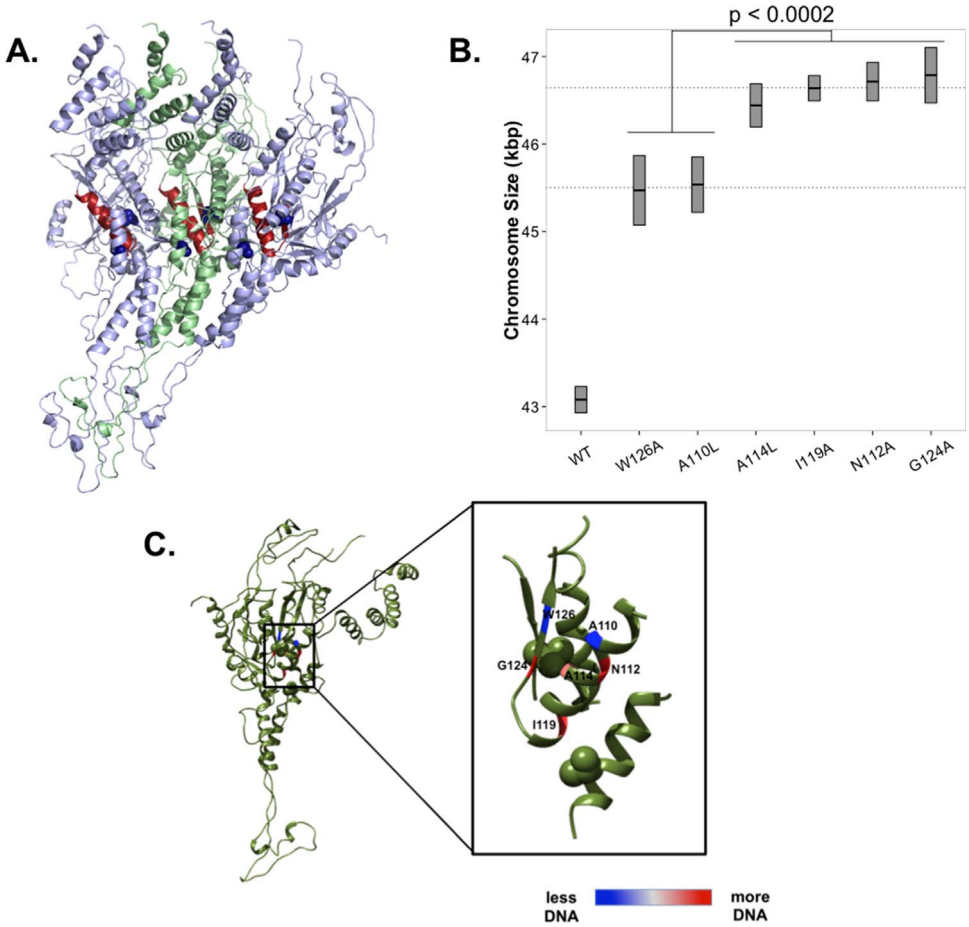

Fig. 5. Chromosome size of mutants in the central structural motif. (A) A trimer of subunits extracted from the dodecameric portal structure. The locations of residues 64 and 303 are highlighted in dark blue, while the structural motif that lays in between 64 and 303 is depicted in red. (B) Plot of chromosome size per mutant arranged in increasing order for the mutants constructed in the central structural motif. These mutants generally result in a more severe overpackaging phenotype than the mutants at positions 64 and 303, and lie within a narrower range of values, speaking to the importance of the portal core in packaging. The mutants appear to cluster around two discrete packaging intervals, in contrast to the behavior observed at positions 64 and 303. (C) Cartoon of the core of the portal protein showing the locations of residues 64 and 303, as well as the central structural motif. The side chains of residues at 64 and 303 are shown as spheres for reference. The locations of the mutants made in the central structural motif are colored by their extent of packaging. It can be seen that the W126A and A110L (blue) lie adjacent to one another in the structure and reside in a slightly less densely packed area of the protein. Conversely, the remaining four mutants are sandwiched between the structural motifs containing residues 64 and 303.

2.5. Mutations that cause overpackaging alter the structural/ dynamic properties of portal

Mutations introduced in the core of the P22 portal protein appear to influence the sensitivity of the packaging termination mechanism to the buildup DNA density. In order to understand the basis of these changes, we biophysically characterized a subset of our mutant portal proteins following recombinant expression and purification. We sought to probe the structural and dynamic properties of these proteins using hydrogen-deuterium exchange mass spectrometry. In these experiments, proteins are incubated in deuterated buffer for a given period of time. Under these conditions, amide hydrogens in the protein backbone are exchanged for deuterium atoms. The extent of deuteration in a given region of the protein is dependent on the intrinsic dynamics and structural features within that region. If the mutations introduced in portal alter the structure or dynamics of the protein, changes in the deuterium uptake should be observed. Shown in Fig. 6A-C are deuterium uptake plots for WT portal and the mutant constructs 64 M and 64 L. In these proteins, only three regions showed significant changes in deuterium uptake between the WT and mutant constructs. More specifically, the two mutant constructs show increased deuteration over the given timespan when compared to WT. This is indicative of increased volume fluctuations in the mutants relative to WT portal. When these peptides are mapped onto the ribbon diagram of the portal structure, as shown in Fig. 6D, it becomes apparent that the observed changes in volume fluctuations are spatially localized. One of the three regions identified, the helix containing residues 58–71 (peptide 2), contains the site of mutation. The other two peptides – 35–44 (peptide 1) and 310–322 (peptide 3) – lie directly in front of this helix. Importantly, 35–44 and 310–322 are spatially located at the junction that connects the portal core to the two stem helices. Changes in the dynamics of this region could readily affect the observed movement of the stem helices between the procapsid and virion forms of the protein, perhaps providing a direct explanation for the packaging phenotypes of these two mutants. Notably, the increase in the exchange rates of the three identified peptides did not result in any compensatory decreases in the exchange rates in other regions of the protein, such as the tunnel loop. These data confirm that the observed packaging phenotype is not due to any previously described effect of the portal protein on packaging.

In their biological context, all studied portal proteins operate as homododecameric complexes. When studied *in vitro*, however, many portal proteins can form structures with varying stoichiometries, including 11-mers, 12-mers, and 13-mers (Lebedev et al., 2007; Sun et al., 2015; Agirrezabala et al., 2005; Trus et al., 2004; Poliakov et al., 2007; Lorenzen et al., 2008). The underlying physical basis for this is the fact that the inter-subunit bonding interactions in these polymorphic structures are mostly unchanged. The similarity in these

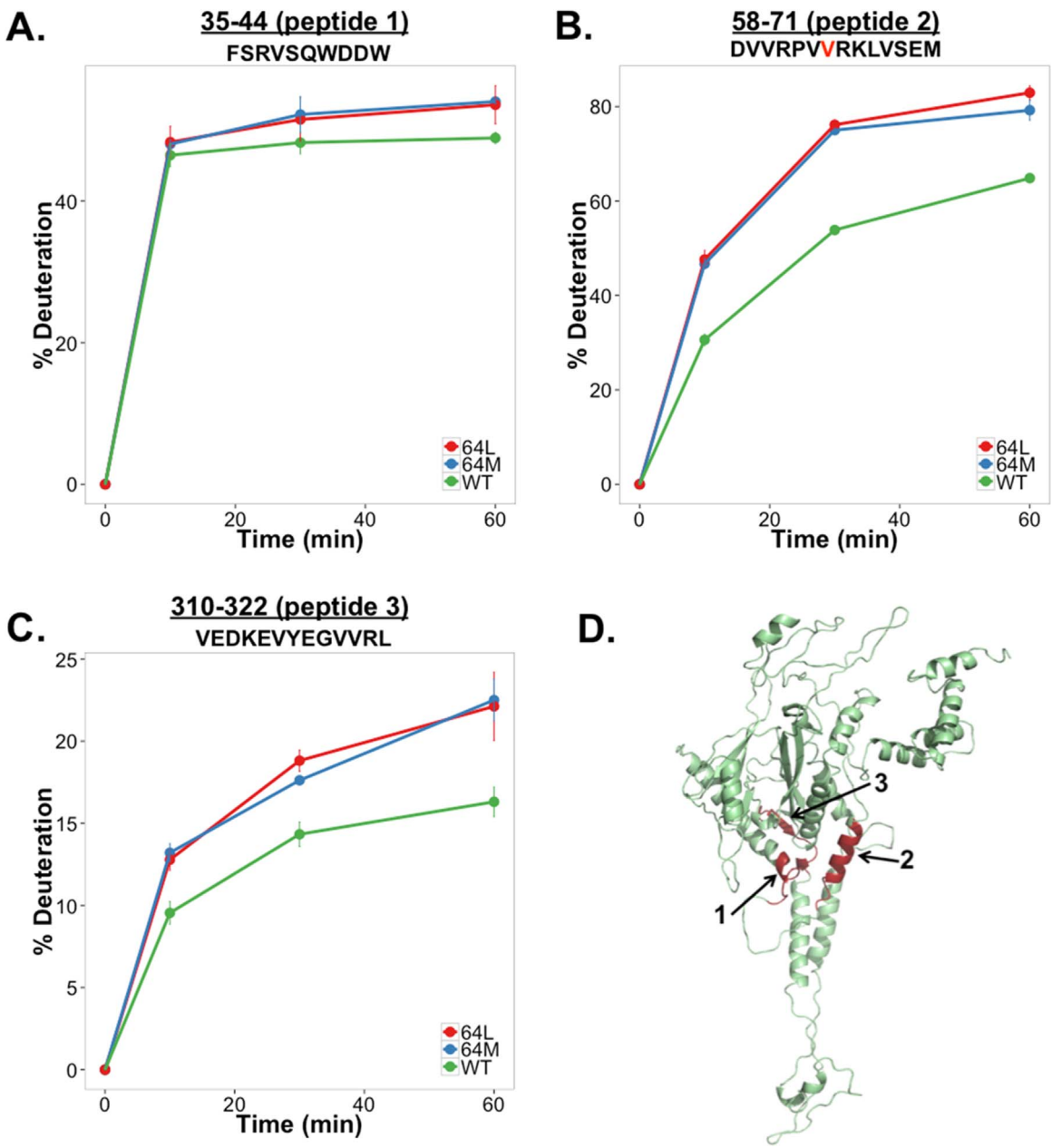

Fig. 6. Hydrogen-deuterium exchange of 64 M and 64 L mutants. (A), (B) and (C) Deuterium uptake plot for peptides 35-44, 58-71, and 310-322, respectively. All of the uptake plots show increased deuteration in the mutants relative to WT over the timescale analyzed. The increase in deuteration is indicative of changes in the volume fluctuations in the mutant proteins relative to WT. Importantly, peptide 58-71 contains the site of the mutation, suggesting that the observed changes are mutant-specific. (D) The location of the three peptides that show changes in deuterium uptake are mapped onto a monomeric portal subunit in red. The helix spanning residues 58-71 is the right most helix in the cartoon. It can be seen that the other two peptides lie directly in front of this α-helix. Importantly, peptides 35-44 and 310-322 are located at the junction between the stem helices and the portal core. Changes in the dynamics of this region could influence the required movement in the stem helices during packaging.

bonding interfaces is due primarily to the complementary electrostatic surfaces that mediate monomer association (Olia et al., 2011; Lebedev et al., 2007; Simpson et al., 2000; Sun et al., 2015). While the portal proteins of SPP1 and T4 can be easily purified from *E. coli* in these various oligomeric states, the WT P22 portal protein is most readily purified in a monomeric form (Olia et al., 2011; Lebedev et al., 2007; Simpson et al., 2000; Sun et al., 2015; Trus et al., 2004; Poliakov et al., 2007; Lorenzen et al., 2008). The position 64 overpackaging mutants mimic this behavior in solution (Fig. 7A). Strikingly, the 110 L, 124 A, 303 M, and 303 T overpackaging mutants form higher order oligomers in solution. Shown in Fig. 7B, the sedimentation coefficient distributions for this group of mutants consist of four predominant peaks with average values of 5.0 (± 0.9) S, 11.2 (± 0.8) S, 17.3 (± 0.6) S, and 25.5 (± 0.4) S, respectively. The 110 L and 124 A mutants predominantly favor the 17.3 S species, whereas the 303 M and 303 T mutants make a mixture of 17.3 and 25.5 S species. This observation suggests location-specific mutant oligomerization preferences. The peaks at 5.0 and 25.5 S likely represent monomeric and dodecameric forms of the portal protein, respectively. This conclusion is based on both the observed s-value for monomeric portal proteins (4.9 S, Fig. 7A) and calculations performed for both the portal monomer and a portal dodecamer with the hydrodynamic modeling program HYDROPRO. In an effort to more accurately assign the peaks at 11.2 and 17.3 S, we imaged the particles by negative stain electron microscopy. Shown in Fig. 8A and B are representative micrographs of our findings. The 303 T sample contains mostly 25.5 S oligomers, and the structures seen in the micrographs have morphologies similar to that of dodecameric portal assemblies. The average diameter of these particles was found to be 18.4 (± 0.8)

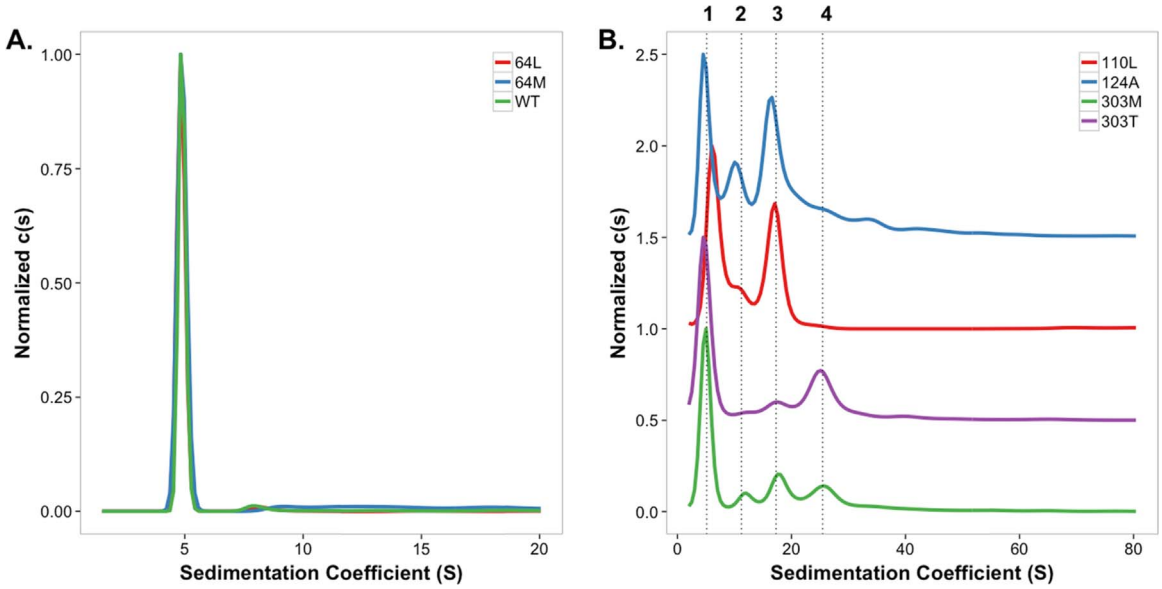

Fig. 7. Examination of the oligomerization state of overpackaging mutants. (A) Sedimentation coefficient distributions of WT, 64M, and 64L portal constructs. All of the samples have measured s-values at around 4.8-4.9 S, corresponding to molecular weight values very close to the expected monomeric mass of 83 kDa. This suggests that all of these constructs remain monomeric in solution. (B) Sedimentation coefficient distributions of 303M, 303T, 110L, and 124A. There are 4 common species in all of the samples, denoted by the vertical dotted lines. These correspond to s-values of 5.0, 11.2, 17.3, and 25.5 S. This is evidence of mutant-specific oligomerization of the portal protein. The peaks at 5.0 and 25.5 S likely correspond to the monomeric and dodecameric forms of the portal protein, respectively, as predicted from both the measured WT s-value and predictions based on hydrodynamic modeling of the portal crystal structure. Assignment of the other peaks is discussed in the text.

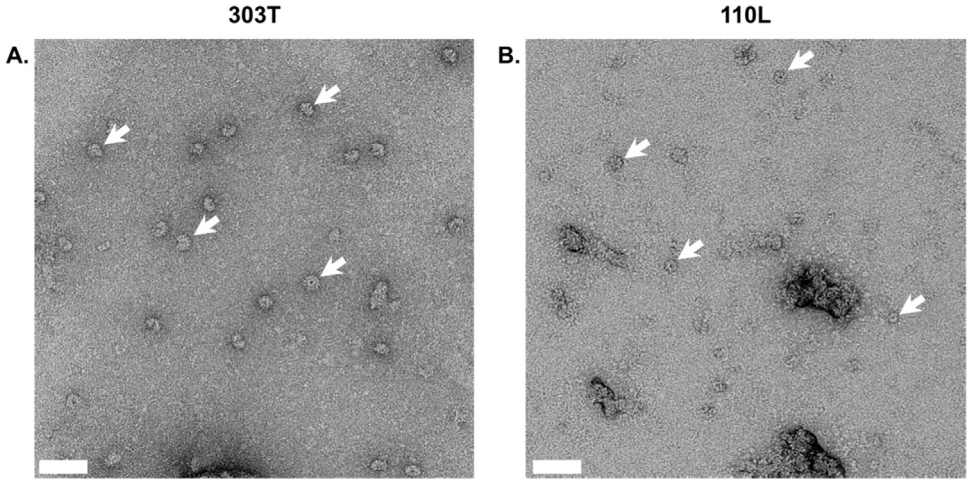

Fig. 8. Electron micrographs of 25.5 and 17.3 S portal species. (A) Representative negative stained electron micrograph of the 25.5 S particles. The particles have an average diameter of approximately 18.5 nm and an overall morphology similar to what would be expected of a dodecameric portal ring. (B) Representative negative stained electron micrograph of the 17.3 S particles. These particles have an average diameter of 16.4 nm, statistically significantly smaller ($p = 0.004$) than that of the 25.5 S particles. The measured size of these particles is consistent with what one might expect for an undecameric portal assembly, consistent with previously biochemical data demonstrating that the P22 portal protein can make undecameric and dodecameric structures in solution. In both micrographs, the scale bar is 100 nm.

nm, close to the average diameter of 18 nm for the two crystallized portal structures. In contrast, in the 110 L sample, which contains mostly 17.3 S assemblies, the morphology of these particles appears to be distinct from those observed in the 303 T sample. The average diameter of these particles was measured to be 16.5 ± 0.9 nm. This is close to the expected diameter of an undecameric portal ring, assuming the size of each portal monomer does not change significantly in the different oligomeric states. Because of this, and because of the ability of the WT P22 portal protein to form undecameric and dodecameric assemblies *in vitro,* we favor the model in which the 17.3 S species is undecameric. Unfortunately, the 11.2 S species is impossible to assign with any certainty.

3. Discussion and conclusions

There is ample evidence pointing to a requirement for portal-containing dsDNA viruses to somehow sense and respond to the density of DNA inside of their capsids (Isidro et al., 2004; Casjens et al., 1992; Feiss et al., 1976). This information on packaging density must be transmitted from the capsid interior to the packaging motor on the capsid exterior for termination of packaging to occur. Recent structural data has suggested a mechanism for this in which the portal dodecamer undergoes significant conformational changes during packaging (Lokareddy et al., 2016). In this model, DNA acts directly on the portal by wrapping itself around the portal perimeter. The grip of the DNA on the portal protein tightens as packaging density increases and this tightening induces long-range conformational changes that alter the terminase binding site at the portal base,

signaling the end of packaging. This model is consistent with data implicating the portal protein as being the sensor for DNA density. However, it does not account for signal transmission throughout the portal itself.

In this study we characterized mutants in the portal protein that change the amount of DNA packaged into virus particles. These mutants lie directly in the core of the protein, away from any important functional domain and likely in the path of signal transmission. The continuous increase in DNA density during packaging is accompanied by a continuous increase in pressure within the virus capsid. The notable difference in internal pressure between the procapsid and the virion suggests that protein compression might be a factor in how the portal protein responds to DNA density. Protein compressibility is an elastic process that is influenced by changes in internal volume caused by alterations in side chain packing (Gekko and Hasegawa, 1986; Maity et al., 2003; Gavish et al., 1983). The introduction of point mutants into a protein alters the side chain packing nearby the site of mutation (Gekko et al., 1996). In turn, this results in mutant-specific changes in the internal volume of the protein and thereby the observed compressibility (Gekko et al., 1996). Comparison of the accessible surface area in a procapsid portal monomer and a virion portal monomer (Fig. 9A) demonstrates that there is a global compression of the P22 portal protein between the procapsid and virion conformations. This implies that protein compression is part of how the portal protein responds to DNA density. The observed lack of correlation between chromosome size and amino acid volume suggests that the overpackaging phenotype is not the result of a steric effect (Gekko et al., 1996). Instead, this observation, along with the continuous distribution of packaging phenotypes, is suggestive of a process influenced by the effects of side chain packing or repacking upon conformational changes. The location-dependent, single-mutant packaging phenotypes observed for mutations at positions 110–126 are also consistent with this model. It can be seen in Fig. 9B that the helix spanning residues 105–121 in the portal core is significantly kinked in the procapsid structure relative to the virion structure. The straightening of this helix, which is likely a consequence of the compression that accompanies DNA packaging must be accompanied by side chain repacking. All of the mutants made within this central region have a more extreme packaging phenotypes than those observed for the 64/303 mutants, suggesting that the central region has a more direct role in signal transmission. The mutants made at positions 112, 114, 119, and 124 have the most pronounced packaging phenotypes. These mutants are more tightly sandwiched between adjacent structural motifs than the mutants at positions 110 or 126 (see Fig. 5C). This is consistent with the interpretation that side chain repacking plays a role in the mechanism of packaging termination, as alterations in the side chain repacking of these residues would have a more pronounced effect on packaging than those in less densely packed positions, i.e. 110 and 126.

Subtle changes in the structure or dynamics of the portal protein, as evidenced by our biophysical characterization of the portal mutants would undoubtedly influence a compression-based information relay. Our observation of mutant-specific oligomerization preferences suggests that the mutations at positions 303, 110, and 124 are likely stabilizing one conformation of the portal protein that facilitates oligomerization relative to the WT protein. Given the overpackaging phenotype, it is likely that these stabilized states more closely resemble the immature conformation of the portal protein. This would make compression of the protein to the mature state require a stronger signal to achieve the same end, thus result in overpackaging. Likewise, the HDX analysis of the position 64 mutants revealed greater volume fluctuations in the mutant constructs relative to WT. Importantly, the observed increase in volume fluctuations in the 64 M and 64 L mutants both occurred at the junction between the portal core and the stem helices. The stem helices in the portal protein change significantly between the procapsid and virion forms of the protein, and have been shown to be important for packaging in SPP1. Therefore, the changes in volume fluctuations observed in the 64 M and 64 L mutants could easily influence the movement of these helices and lead to the observed changes in the packaging phenotype.

4. Conclusions

Mutations in the central region of the P22 portal protein result in pronounced changes in the amount of DNA packaged. This region is not involved in interactions with packaged DNA or terminase binding. Nonetheless, it has a pronounced affect on chromosome length. The difference in accessible surface area between the procapsid and virion forms of the portal protein suggests that there is an overall compres-

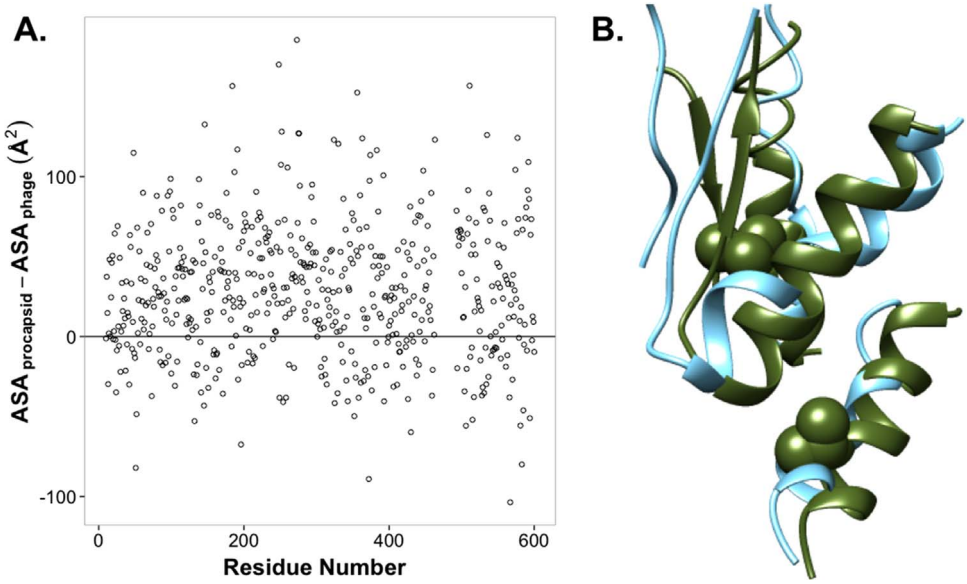

Fig. 9. Evidence of protein compression between portal conformations. (A) Plot of the difference in the solvent accessible surface area (ASA) between the procapsid and phage forms of the portal protein. The net positive values demonstrate that there is compression between the procapsid and phage conformations of the protein. (B) An alignment of the cores of the two forms of the portal protein. The procapsid form, shown in blue, shows a less tightly packed conformation relative to the phage form, shown in green. Additionally, the central helix (residues 105-121) is significantly kinked in the procapsid form relative to the phage form. The straightening of this helix, along with the compression of the protein, are likely major determinants of the long-range signaling mechanism between the crown and stalk regions of portal.

sion of the protein in transitioning between the procapsid and virion forms. Characterization of the packaging phenotypes of the portal mutants revealed phenotype characteristics consistent with the hypothesis that overpackaging is caused by alterations in protein compression. This hypothesis was further corroborated by biophysical measurements suggesting that the introduced mutations alter the volume fluctuations of the portal protein. These results are consistent with a model in which increasing DNA density induces compression of the portal protein. This pressure-sensitive signal is the means by which the portal protein relays information on packaging density to the terminase complex on the capsid exterior. Alterations in the compressibility of the portal protein appear to result in a less sensitive switch, but do not fundamentally alter the mechanism of signal transmission.

5. Methods

5.1. Cloning and mutagenesis

The desired mutants were engineered into the portal protein in a two-step PCR procedure. Construction of the WT gene in the pNFPW vector was described previously. The pNFPW plasmid contains a lacUV5 promoter upstream of the cloning site for inducible expression of the gene of interest. All of the mutants were constructed off of that plasmid. Briefly, two end primers were synthesized and used in all PCR steps. The 5′ forward primer containing a *Kpn*I restriction site and a canonical ribosome binding site had the sequence TTAA GGTACCAAGGAGATATAATG. The 3′ reverse primer containing an *Xba*I restriction site had the sequence AATTTCTAGATTATTGA GGTGTCTCGGCTA. In two separate reactions, one of two complementary, single stranded primers containing the mutation of interest were put into PCR reactions with either the forward or reverse end primer. This process yields two gene fragments, both with the mutation of interest. The two gene fragments were gel purified and combined into a single reaction with both end primers to generate the entire 2.2 kb mutant gene. The mutant gene was gel purified and digested with *Kpn*I and XbaI restriction enzymes and ligated into the pNFPW backbone. This plasmid was then transformed into the *Salmonella* strain DB7000 for transcomplementation.

5.2. Transcomplementation

Transcomplementation experiments were carried out as previously described. Briefly, overnight cultures of DB7000 containing the plasmids of interest were grown up in LB containing ampicillin. The overnight cultures were diluted 1:100 into fresh media and allowed to grow to 3×10^8 cells/mL. The cells were simultaneously infected with *1⁻amE5*13⁻amH101* P22 at an MOI of 7 and induced for portal expression with 0.075 mM IPTG. This strain of P22 harbors amber mutations in genes *1* and *13*, which occlude portal production from the chromosomal gene and prevent cell lysis, respectively. Infection and induction were carried out for 1 h, at which point cells were lysed with chloroform, cell debris was cleared, and additional handling could be carried out in an experiment-dependent manner. For titering experiments, dilutions were made directly from the cleared media. Permissive titers were obtained on the *Salmonella* strain MS1363. For all other experiments involving phage, phage were precipitated in 10% PEG 8000 and 1 M NaCl. The precipitate was pelleted at 18,600×*g* for 1 h and resuspended in 20 mL phage buffer (50 mM Tris pH 7.6, 100 mM NaCl, 5 mM $MgCl_2$). P22 particles were pelleted again by centrifugation at 165,000×*g* for 40 min. This pellet was resuspended in 1 mL of the same buffer. Insoluble material was cleared by centrifugation at 17,000×*g* for 30 min and the supernatant was stored at 4 °C until use.

5.3. DNA purification

For DNA purification, the stored phage supernatant was pelleted at 136,000×*g* for 15 min, and resuspended in 400 μL 10 mM Tris pH 8.0, 5 mM $MgCl_2$. After resuspension, the following reagents were added in order to final concentrations: 100 mM Tris pH 8.0, 0.2% SDS, 20 mM EDTA. The solution was this incubated at 70 °C for 15 min. Potassium acetate was then added to a final concentration of 450 mM, the solution was mixed and then incubated on ice for 1 h. This step leads to the precipitation of proteinaceous material from solution. The precipitate was cleared by centrifugation for 30 min at 4 °C at 2 volumes of ice-cold ethanol were added to the supernatant. The precipitated DNA was gently spun around a glass capillary and allowed to air dry. The dried DNA was washed with 70% ethanol and resuspended in 500 μL of DNA sizing buffer (10 mM Tris pH 8.0, 100 mM NaCl, 1 mM EDTA).

5.4. Analytical ultracentrifugation of DNA

For analytical ultracentrifugation, the purified DNA was diluted to 0.005 wt% in DNA sizing buffer. This dilute DNA concentration is important to prevent the occurrence of the Johnston-Ogsten effect during the sedimentation experiments. Boundary migration was monitored by absorbance at 260 nm. The samples were spun in epon double-sectored centerpieces (Spin Analytical) in an An-60 Ti rotor at 15,000 rpm and 25 °C (Beckmann). Scans were collected every 4 min. The data were analyzed with the ls-g*(s) algorithm in SEDFIT (www.analyticalultracentrifugation.com). The resulting peak in each s-value distribution was integrated in order to obtain the weight-average s-value in a given run for each sample. This value was converted to $s_{w,20}$ for application of the scaling law. The distribution widths at 1σ were calculated in SEDPHAT. Briefly, all three data sets for a given mutant are globally analyzed in SEDPHAT with the average s-value and calculated mass fixed and the column meniscus and base allowed to float. Following minimization, the input s-value was slightly increased or decreased and the fit was performed again. The new chi-squared value was compared to the critical chi-squared calculated in SEDPHAT using F-statistics. Once the new chi-squared was greater than the critical value, the last input s-value was recorded as the upper or lower boundary at 1σ.

5.5. Cryo-electron microscopy

Particles to be imaged for cryo-EM were pelleted for 20 min at 136,000 x g and resuspended in 50 μL of 50 mM Tris pH 7.6, 50 mM NaCl, 5 mM $MgCl_2$. For imaging, 2-fold dilutions were prepared in the same buffer and the sample was applied to carbon holey-film grids and flash frozen in liquid ethane. The micrographs were collected in an FEI Tecnai F20 electron microscope operated at 200 kV with a Gatan UltraScan 4000 CCD camera. Images were collected at magnifications of 29,000x or 50,000x with defocus values ranging from 2 to 3 μm. The electron dose was around 20 e/Å^2. The images were analyzed in ImageJ (https://imagej.nih.gov/ij/). For both samples, just over 200 particles were included in the particle-type distribution analysis. Of these 200 particles, packaged virions constituted 78 and 74 particles, in the WT and 64 L samples, respectively. The diameter of these particles were measured in ImageJ relative to the scale bar present on each micrograph.

5.6. Recombinant protein expression and purification

The constructs used for recombinant expression of select portal mutants were created using a PCR strategy similar to the one described above. Briefly, a 5′ end primer with an *Xba*I restriction site and a 3′ end primer with a XhoI restriction site were used along with the same complementary mutant primers described above to synthesize two gene fragments containing the desired mutation. These gene fragments

were then used to amplify the whole portal gene. The full-gene PCR product was then digested with *Xba*I and *Xho*I and ligated into a pET21a backbone. These plasmids were introduced into BL21(DE3) *E. coli* for overexpression. Overnight cultures of these constructs were grown in LB media with 100 μg/mL ampicillin. These cultures were diluted 1:100 into fresh media the following morning and grown to mid-log phase. At mid-log phase, the cultures were induced with 1 mM IPTG for 5 h at 25 °C. Cells were pelleted and concentrated 7.5x in cold breakage buffer – 50 mM Tris pH 7.6; 50 mM NaCl; 2 mM EDTA; 0.05% Tween20; 0.01% Tween 80 – with two tablets of EDTA-free protease inhibitor (Pierce), 2 mM PMSF, and 0.2 mg/mL lysozyme. The resuspended cells were frozen at −80 °C until thawing and lysis. Lysis was carried out by freeze-thaw followed by sonication to reduce viscosity. Cell debris was pelleted by centrifugation at 18,000×*g* for one hour. Ammonium sulfate was added to the supernatant to 36% saturation and stirred at 4 °C overnight. The resulting precipitate was pelleted by centrifugation at 18,000×*g* for one hour and resuspended in an equal amount of breakage buffer. The resuspension was then loaded onto a 5 mL HiTrap Q HP column (GE Life Sciences) and washed in LS buffer, breakage buffer without any detergent included. Portal was eluted in a linear, twenty-five minute gradient flowing at 2 mL/min from LS buffer to HS buffer, containing 1 M NaCl and otherwise equivalent buffer conditions. Portal consistently eluted at approximately 300–400 mM NaCl. The portal-containing fractions were pooled, concentrated, and loaded onto a HiPrep 16/60 Sephacryl S-200 gel filtration column (GE Life Sciences) equilibrated in LS buffer. The peak portal-containing fractions were pooled and stored at −80 °C until use.

5.7. Analytical ultracentrifugation of proteins

Protein samples for AUC were prepared by concentrating the proteins to 2 mg/mL by spin concentration, incubation at RT for approximately 1 h, followed by subsequent dilution in LS buffer to 0.5 mg/mL for analysis. The same hardware was used for these experiments as for the experiments analyzing chromosome size. Monomeric proteins were spun at 35,000 rpm and 25 °C. Samples with oligomers were spun at 18,000 rpm and 25 °C. 150 total scans were collected per sample with a sampling rate of 1 scan every 4 min. The resulting data was analyzed using the c(s) model in SEDFIT with the input parameters of vbar =0.73 mL/g; density =1.0009*g*/mL; viscosity =0.0091043 Poise.

5.8. Negative stain electron microscopy

The same samples prepared for AUC were used in imaging the oligomeric species. The microscope set up with similar to that used for cryo-EM. 20x dilutions of protein samples were applied to carbon grids and stained with 2% uranyl formate. Particles were imaged at 65,500x. Oligomer dimensions were quantified using ImageJ software similarly to the analysis of whole virus particles from cryo-EM. Measurements were made on 30 particles from both samples.

5.9. Hydrogen-deuterium exchange

For hydrogen-deuterium exchange experiments, proteins were exchanged in a 90% D_2O buffer containing 50 mM Tris pH 7.5 and 50 mM NaCl. The proteins were concentrated to 5 mg/mL and 3 μL was added to a tube equilibrated at 25 °C. 50 μL of deuterated buffer was then spiked on top of the protein solution to facilitate rapid mixing and the sample was allowed to incubate for the desired timespan. Quenching of the reaction was achieved by rapidly adding 50 μL of 150 mM sodium phosphate pH 2.5; 1 M urea in 0.1% formic acid. Following quenching, the samples were immediately frozen in liquid nitrogen and stored at −80 °C until injection.

For sample analysis, each sample was thawed individually just before injection. Concentrated pepsin was spiked into the frozen sample to a final concentration of 10 mg/mL and 50 μL of the slushy solution was injection into an ice cold loading loop. Digestion was allowed to proceed for exactly 2 min before ejection onto a Phenomenex Aeris 3.6 μm XB-C18 reverse-phase column for peptide separation. Peptides were separated with a linear ten-minute gradient from 5% to 95% acetonitrile in 0.1% formic acid. The column eluted into a Q-ToF Premier mass spectrometer (Waters) and peptides were fragmented with MSe data acquisition settings. From the resulting mass spectra, the peptides were identified using ProteinLynx Global Server (Waters). The spectra from the exchanged samples were input into HDExaminer (Sierra Analytics) to calculate % Deuteration for each peptide at each time point.

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